16. Scenario: You are working on a project that involves analyzing customer reviews for a product. You have a dataset containing customer reviews, and your task is to develop a Python program that calculates the frequency distribution of words in the reviews.

Question: Develop a Python program to calculate the frequency distribution of words in the customer reviews dataset?

```
from collections import Counter
import re

reviews = [
    "The product is good",
    "The product is excellent",
    "Good product and good quality"
]

text = " ".join(reviews).lower()
words = re.findall(r'\b\w+\b', text)

frequency = Counter(words)

print("Word Frequency Distribution:")
for word, count in frequency.items():
    print(word, ":", count)
```

```
========== RESTART: C:/Us
Word Frequency Distribution:
the : 2
product : 3
is : 2
good : 3
excellent : 1
and : 1
quality : 1
```

17. Scenario: You are a data analyst working for a marketing research company. Your team has collected a large dataset containing customer feedback from various social

media platforms. The dataset consists of thousands of text entries, and your task is to develop a Python program to analyze the frequency distribution of words in this dataset. Your program should be able to perform the following tasks:

 Load the dataset from a CSV file (data.csv) containing a single column named "feedback"

with each row representing a customer comment.

 Preprocess the text data by removing punctuation, converting all text to lowercase, and

eliminating any stop words (common words like "the," "and," "is," etc. that don't carry

significant meaning).

 Calculate the frequency distribution of words in the preprocessed dataset.

 Display the top N most frequent words and their corresponding frequencies, where N is

provided as user input.

 Plot a bar graph to visualize the top N most frequent words and their frequencies.

Question: Create a Python program that fulfills these requirements and helps your team gain

insights from the customer feedback data.

```
import pandas as pd
import matplotlib.pyplot as plt
import re
from collections import Counter

# Load CSV file
df = pd.read_csv("data.csv")

# Stop words
stop_words = {
    "the", "and", "is", "a", "an", "of", "to",
```

```
    "in", "it", "for", "was", "are", "this", "that"
}

# Preprocess feedback
words = []
for feedback in df["feedback"].dropna():
    text = re.sub(r'[^\w\s]', '', str(feedback).lower())
    words.extend(
        word for word in text.split()
        if word not in stop_words
    )

# Calculate word frequencies
frequency = Counter(words)

# Get N from user
n = int(input("Enter the number of top words: "))
top_words = frequency.most_common(n)

print("Top frequent words:")
for word, count in top_words:
    print(word, ":", count)

# Plot bar graph
if top_words:
    labels, values = zip(*top_words)
    plt.bar(labels, values)
    plt.title("Top Frequent Words in Customer Feedback")
```

```
plt.xlabel("Words")
plt.ylabel("Frequency")
plt.xticks(rotation=45)
plt.tight_layout()
plt.show()
```

18. Suppose a hospital tested the age and body fat data for 18 randomly selected adults with the

following result.

age	23	23	27	27	39	41	47	49	50
%fat	9.5	26.5	7.8	17.8	31.4	25.9	27.4	27.2	31.2
age	52	54	54	56	57	58	58	60	61
%fat	34.6	42.5	28.8	33.4	30.2	34.1	32.9	41.2	35.7

Question:

 Calculate the mean, median and standard deviation of age and %fat using Pandas.

 Draw the boxplots for age and %fat.

 Draw a scatter plot and a q-q plot based on these two variables

```
import pandas as pd
import matplotlib.pyplot as plt
from scipy import stats

# Dataset from the given table
df = pd.DataFrame({
    'Age': [23, 23, 27, 27, 39, 41, 47, 49, 50,
            52, 54, 54, 56, 57, 58, 58, 60, 61],
    '%Fat': [9.5, 26.5, 7.8, 17.8, 31.4, 25.9,
```

```
        27.4, 27.2, 31.2, 34.6, 42.5, 28.8,
        33.4, 30.2, 34.1, 32.9, 41.2, 35.7]
})

# Statistical measures
print("Mean:\n", df.mean())
print("\nMedian:\n", df.median())
print("\nStandard Deviation:\n", df.std())

# Boxplots
df[['Age', '%Fat']].plot(
   kind='box', subplots=True, layout=(1, 2),
   figsize=(8, 4), legend=False
)
plt.tight_layout()
plt.show()

# Scatter plot
plt.scatter(df['Age'], df['%Fat'])
plt.title("Age vs Body Fat")
plt.xlabel("Age")
plt.ylabel("Body Fat (%)")
plt.show()

# Q-Q plots
fig, axes = plt.subplots(1, 2, figsize=(10, 4))
stats.probplot(df['Age'], dist="norm", plot=axes[0])
axes[0].set_title("Q-Q Plot of Age")
```

```
stats.probplot(df['%Fat'], dist="norm", plot=axes[1])

axes[1].set_title("Q-Q Plot of Body Fat")

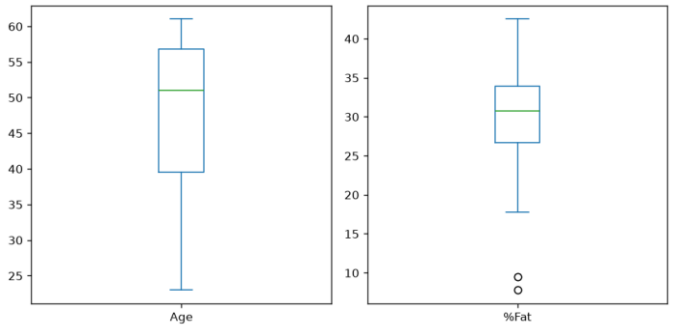

plt.tight_layout()

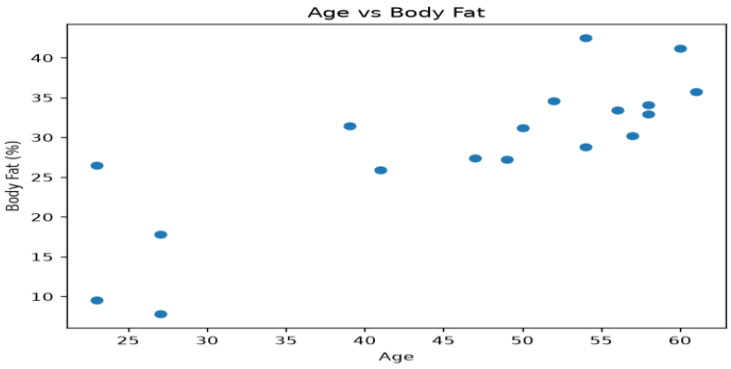
plt.show()
```

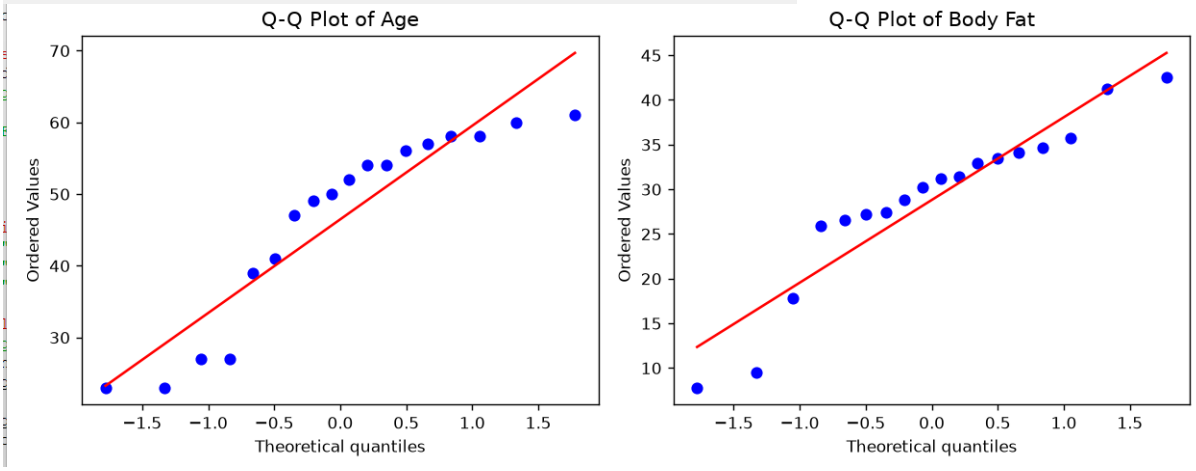
```
Mean:
 Age      46.444444
%Fat     28.783333
dtype: float64

Median:
 Age      51.0
%Fat     30.7
dtype: float64

Standard Deviation:
 Age      13.218624
%Fat      9.254395
dtype: float64
```







19. Scenario:

You are a medical researcher investigating the effectiveness of a new drug in reducing blood pressure. You conduct a clinical trial with a sample of 50 patients who were randomly assigned to receive either the new drug or a placebo. After measuring their blood pressure levels at the end of the trial, you obtain the data for both groups. Now, you want to determine the confidence intervals for the mean reduction in blood pressure for both the drug and placebo groups.

```
import numpy as np
from scipy import stats

# Illustrative data: replace with actual trial measurements
drug = np.array([
    12, 15, 10, 14, 13, 16, 11, 15, 14, 12,
    17, 13, 16, 14, 11, 15, 12, 14, 16, 13,
    10, 15, 14, 12, 16
])

placebo = np.array([
    5, 7, 4, 6, 8, 5, 3, 7, 6, 4,
    8, 5, 6, 7, 4, 5, 9, 6, 3, 7,
    5, 4, 6, 8, 5
])

# Calculate 95% confidence intervals
for name, data in [('Drug', drug), ('Placebo', placebo)]:
    mean = np.mean(data)
    ci = stats.t.interval(
        0.95,
```

```
    df=len(data) - 1,
    loc=mean,
    scale=stats.sem(data)
  )
  print(name, "Mean Reduction:", round(mean, 2))
  print(name, "95% Confidence Interval:",
      (round(ci[0], 2), round(ci[1], 2)))
```

```
Drug Mean Reduction: 13.6
Drug 95% Confidence Interval: (np.float64(12.78), np.float64(14.42))
Placebo Mean Reduction: 5.72
Placebo 95% Confidence Interval: (np.float64(5.05), np.float64(6.39))
```

Question:

"What is the 95% confidence interval for the mean reduction in blood pressure for patients who received the new drug? Also, what is the 95% confidence interval for the mean reduction in blood pressure for patients who received the placebo?

20. Scenario:

You are a data scientist working for an e-commerce company. The marketing team has conducted an A/B test to evaluate the effectiveness of two different website designs (A and B) in terms of conversion rate. They randomly divided the website visitors into two groups, with one group experiencing design A and the other experiencing design B. After a week of data collection, you now have the conversion rate data for both groups. You want to determine whether there is a statistically significant difference in the mean conversion rates between the two website designs.

Question:

"Based on the data collected from the A/B test, is there a statistically significant difference in the mean conversion rates between website design A and website design B?"

```
import numpy as np
from scipy import stats
# Illustrative data: 1 = conversion, 0 = no conversion
design_A = np.array([
```

```
    1, 0, 1, 1, 0, 1, 0, 1, 1, 0,
    1, 0, 1, 1, 0, 1, 0, 1, 0, 1
])
design_B = np.array([
    1, 1, 1, 0, 1, 1, 0, 1, 1, 1,
    0, 1, 1, 0, 1, 1, 1, 0, 1, 1
])
# Calculate conversion rates
rate_A = np.mean(design_A) * 100
rate_B = np.mean(design_B) * 100
# Perform independent two-sample t-test
t_stat, p_value = stats.ttest_ind(
    design_A, design_B, equal_var=False
)
print("Design A Conversion Rate:", rate_A, "%")
print("Design B Conversion Rate:", rate_B, "%")
print("P-value:", round(p_value, 4))

if p_value < 0.05:
    print("Statistically significant difference")
else:
    print("No statistically significant difference")
```

```
Design A Conversion Rate: 60.0 %
Design B Conversion Rate: 75.0 %
P-value: 0.3237
No statistically significant difference
```